Ch 5.1.3-4: k-Fold Cross-Validation

Lecture 13 - CMSE 381

Prof. Elizabeth Munch

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Wed, Oct 2, 2024

Announcements

Last time:

- Validation Set
- LOOCV

Announcements:

- Exam 1 grades
- HW #4 Posted.
  - Changed Deadline! Due Wednesday Oct 9.

Lec #	Date			Reading	HW
12	Mon	9/30	Leave one out CV	5.1.1, 5.1.2	
13	Wed	10/2	k-fold CV	5.1.3	
14	Fri	10/4	More k-fold CV,	5.1.4-5	
15	Mon	10/7	k-fold CV for classification	5.1.5	
16	Wed	10/9	Resampling methods: Bootstrap	5.2	HW #4 Due Weds 10/9
17	Fri	10/11	Subset selection	6.1	
18	Mon	10/14	Shrinkage: Ridge	6.2.1	
19	Wed	10/16	Shrinkage: Lasso	6.2.2	
20	Fri	10/18	Dimension Reduction	6.3	HW #5 Due Fri 10/18
	Mon	10/21	No class - Fall break		
	Wed	10/23	**Review**		
	Fri	10/25	**Midterm #2**		

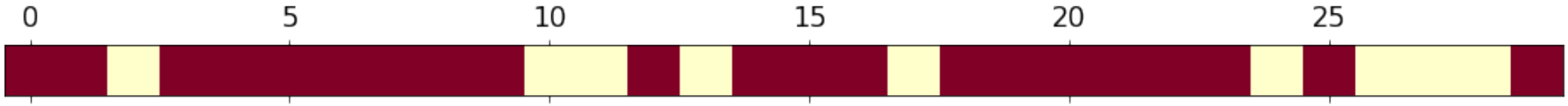
Covered in this lecture

- k-fold CV

Section 1

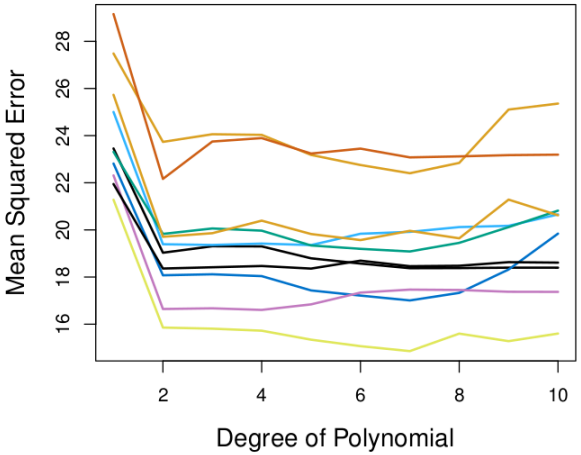
Last time

Validation set approach

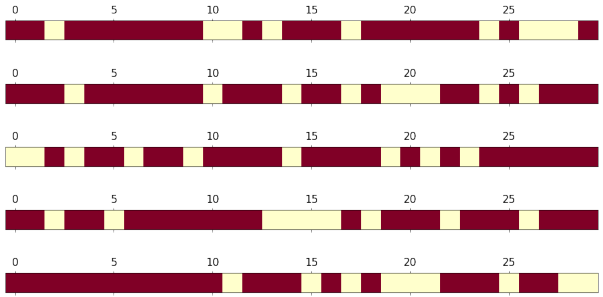


- Divide randomly into two parts:
  - Training set
  - Validation/Hold-out/Testing set
- Fit model on training set
- Use fitted model to predict response for observations in the test set
- Evaluate quality (e.g. MSE)

Problems



Ex. Predict `mpg` using `horsepower`



- Highly variable results, no consensus about the error
- Tends to overestimate test error rate

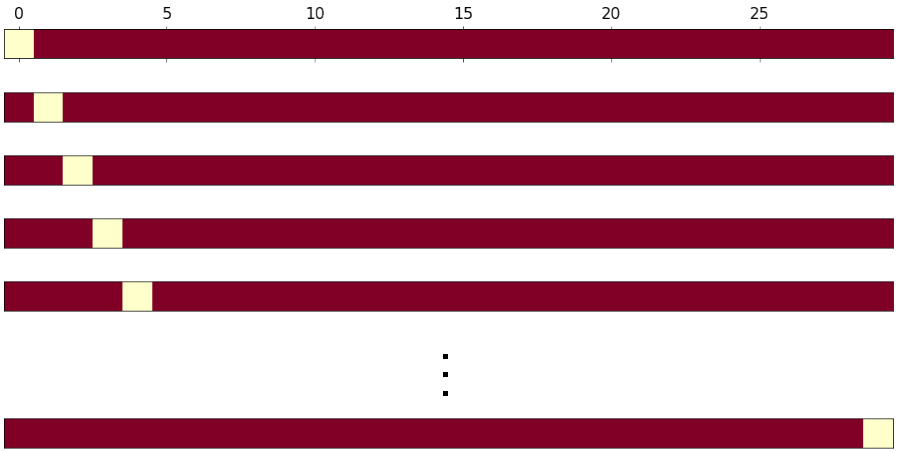
Leave One Out CV (LOOCV)

cross validation

- Remove (x_1, y_1) for testing.
- Train the model on $n-1$ points: $\{(x_2, y_2), \cdots, (x_n, y_n)\}$
- Calculate $\mathrm{MSE}_1 = (y_1 - \hat{y}_1)^2$

- Remove (x_2, y_2) for testing.
- Train the model on $n-1$ points: $\{(x_1, y_1), (x_3, y_3), \cdots, (x_n, y_n)\}$
- Calculate $\mathrm{MSE}_2 = (y_2 - \hat{y}_2)^2$

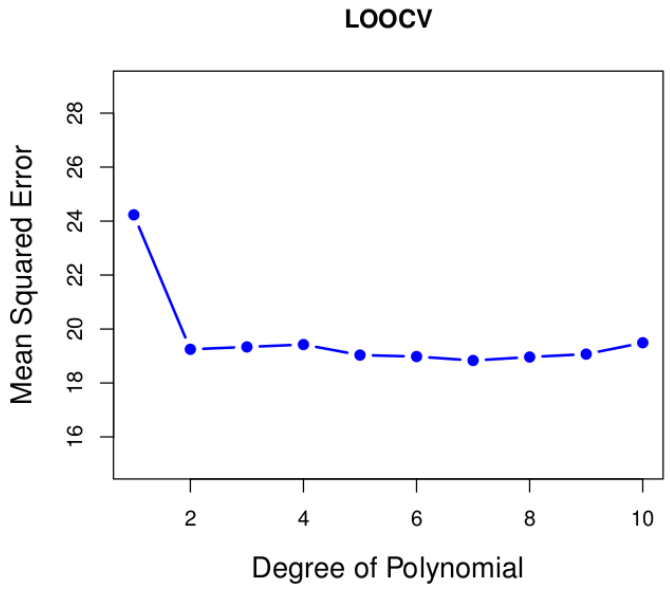
- Rinse and repeat



Return the score:

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^{n} \mathrm{MSE}_i$$

Pros and Cons



- No variance
- Higher computation cost

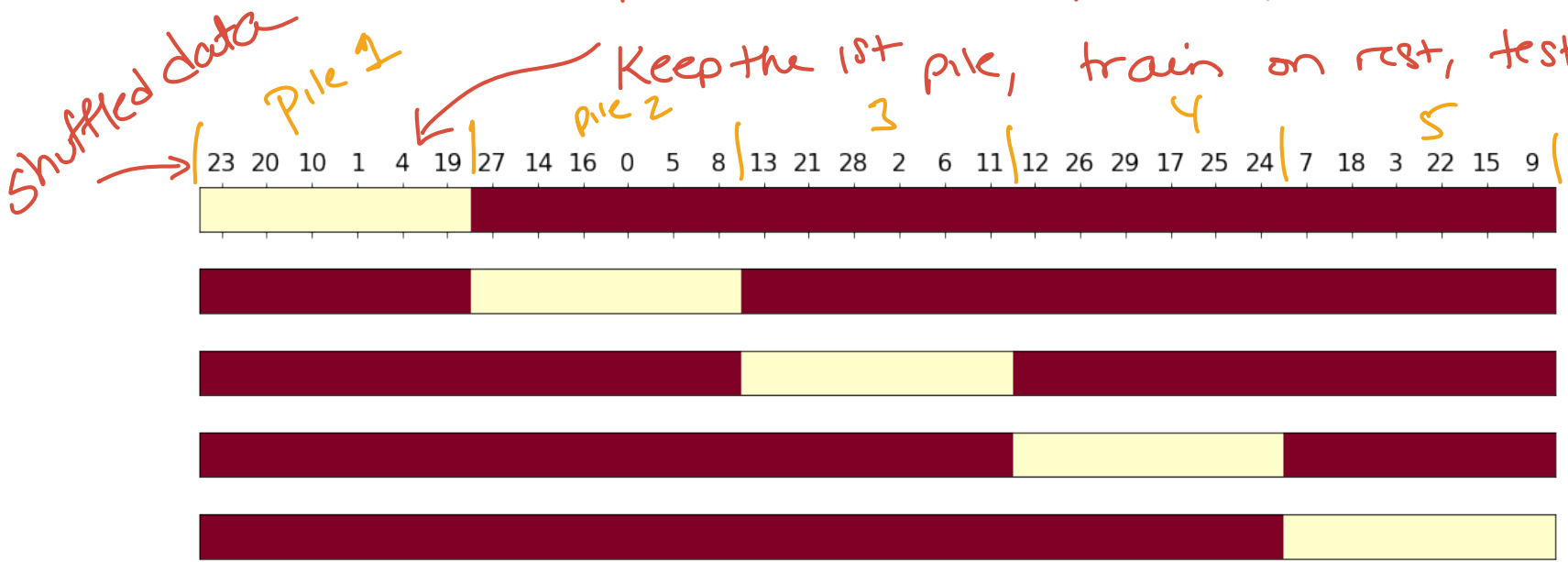
Section 2

k-Fold CV

The idea

K = 5 different piles of data points

Keep the 1st pile, train on rest, test on 1st pile



If you make k = n = # of data points

this is LOOCV

Mathy version

- Randomly split data into k-groups (folds) ← my "pile" of data
- Approximately equal sized. For the sake of notation, say each set has ℓ points

  my total # of pts
  $n = \ell \cdot k$

- Remove ith fold U_i and reserve for testing. ← "the ith pile"
- Train the model on remaining points
- Calculate $\mathrm{MSE}_i = \frac{1}{\ell} \sum_{(x_j, y_j) \in U_i} (y_j - \hat{y}_j)^2$

  ↑ # of data pts in test set

- Rinse and repeat

Return

$$CV_{(k)} = \frac{1}{k} \sum_{i=1}^{k} \mathrm{MSE}_i$$

k-fold cross validation score

By hand first!

There are 10 students in the class, and we have data points for each. They have already been randomly permuted below. Write down the training/testing sets for a 3-fold CV

- Damien
- Alice
- Greta
- Jasmin

Fold 1 U_1

- Benji
- Inigo
- Firas

Fold 2 U_2

- Carina
- Enrique
- Hubert

Fold 3 U_3

Fold 1	**Fold 2**	**Fold 3**
Test U_1	Test U_2	Test U_3
D, A, G, J	B I F	C E H
Training	Training	Training
B I F C E H	D A G J C E H	D A G J B I F

Coding - Building k-fold CV

Pros and Cons

Pros:

Cheaper computation than LOOCV

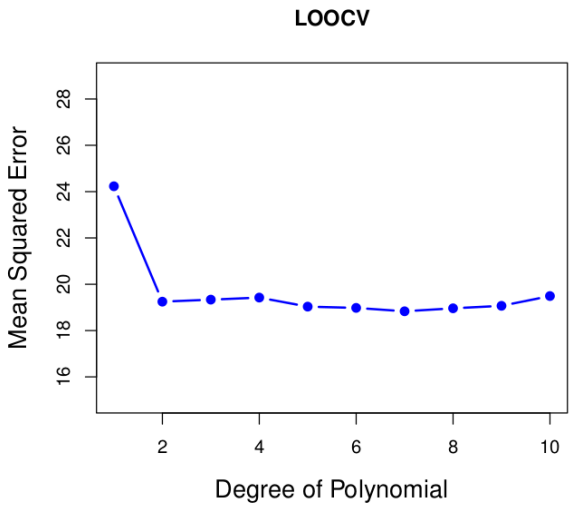
Common $k = 5, 10$

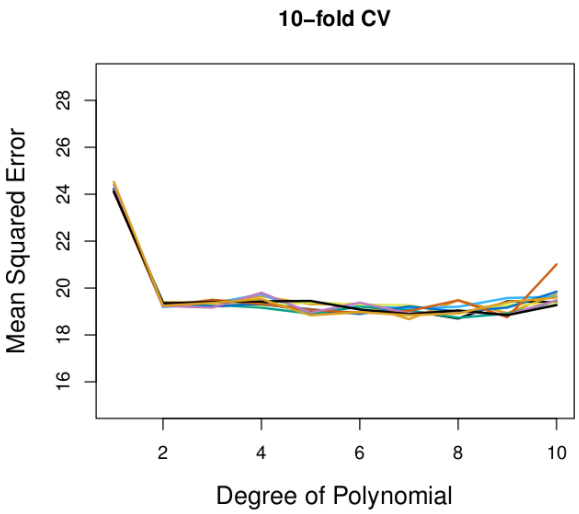
Cons:

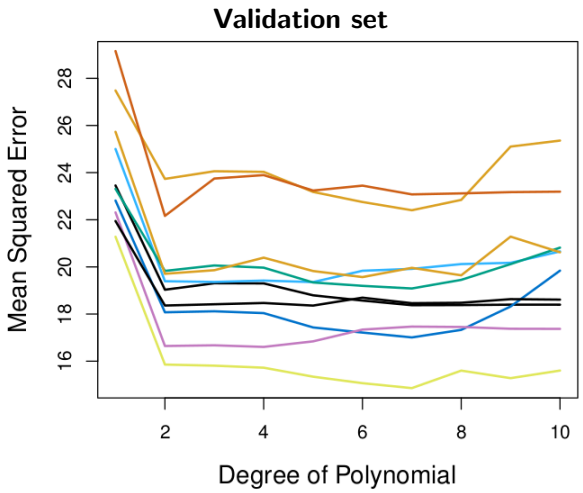
Unlike LOOCV has randomness

Less variability than validation set

Comparison


LOOCV
Mean Squared Error
16
18
20
22
24
26
28
2
4
6
8
10
Degree of Polynomial


10–fold CV
Mean Squared Error
16
18
20
22
24
26
28
2
4
6
8
10
Degree of Polynomial


Validation set
Mean Squared Error
16
18
20
22
24
26
28
2
4
6
8
10
Degree of Polynomial

Next time

Lec #	Date			Reading	HW
12	Mon	9/30	Leave one out CV	5.1.1, 5.1.2	
13	Wed	10/2	k-fold CV	5.1.3	
14	Fri	10/4	More k-fold CV,	5.1.4-5	
15	Mon	10/7	k-fold CV for classification	5.1.5	
16	Wed	10/9	Resampling methods: Bootstrap	5.2	HW #4 Due Weds 10/9
17	Fri	10/11	Subset selection	6.1	
18	Mon	10/14	Shrinkage: Ridge	6.2.1	
19	Wed	10/16	Shrinkage: Lasso	6.2.2	
20	Fri	10/18	Dimension Reduction	6.3	HW #5 Due Fri 10/18
	Mon	10/21	No class - Fall break		
	Wed	10/23	**Review**		
	Fri	10/25	**Midterm #2**		